



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

CHI PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 Q&A on lightweight routing, middleware, context, and net/http

TOTAL: 10 QUESTIONS

Q1 What is Chi and why is it described as idiomatic Go? [Threat Model](#)

Q6 How do you implement JWT authentication middleware in Chi? [Threat Model](#)

Q2 How does Chi middleware chaining work? [Architecture](#)

Q7 What built-in middleware does Chi provide? [Lifecycle](#)

Q3 How do you read URL parameters in Chi? [Configuration](#)

Q8 How do you define RESTful resource routes in Chi? [Compliance](#)

Q4 What is chi.Mount() and how do you use it? [Hardening](#)

Q9 How does Chi handle 404 and 405 responses? [Testing](#)

Q5 How do you pass values between middleware and handlers in Chi? [Threat Model](#)

Q10 How do you write tests for Chi handlers? [Tuning](#)



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Chi is a lightweight router that implements

Mitigation

Chi is a lightweight router that implements `http.Handler` and uses only standard `net/http` no custom context types. Middleware is standard `func(http.Handler) http.Handler`. Any existing `net/http` handler works out of the box without adaptation.

A6 Write a `func(next http.Handler) http.Handler` that reads

Resiliability

Write a `func(next http.Handler) http.Handler` that reads the `Authorization` header, validates the JWT, and on success calls `next.ServeHTTP(w, r.WithContext(ctx))` with the user in the context. On failure, write 401 and return without calling next.

A2 Chi uses alice-style functional composition.

Primitives

Chi uses alice-style functional composition. `r.Use(middleware.Logger)` applies globally. `r.Group(func(r chi.Router) { r.Use(AuthMiddleware); r.Get('/admin', handler) })` scopes middleware to the group. Each middleware wraps the next `http.Handler`.

A7 `middleware.Logger` (structured request logging),

Automation

`middleware.Logger` (structured request logging), `middleware.Recoverer` (panic recovery), `middleware.Throttle(N)` (concurrent request limiter), `middleware.Compress` (gzip), `middleware.StripSlashes`, `middleware.Heartbeat('/health')`, and `middleware.RealIP`.

A3 `chi.URLParam(r, 'id')` reads the named parameter from

Hardening

`chi.URLParam(r, 'id')` reads the named parameter from the URL. In a handler: `id := chi.URLParam(r, 'id')`. Chi stores params in the request context under its own key, making them accessible without a custom Context type via the standard library.

A8 `r.Get('/users', listUsers); r.Post('/users', createUser);`

Governance

`r.Get('/users', listUsers); r.Post('/users', createUser); r.Route('/users/{id}', func(r chi.Router) { r.Get('/', getUser); r.Put('/', updateUser); r.Delete('/', deleteUser) })`. Chi does not have a `resources()` shortcut explicit routes are idiomatic.

A4 `r.Mount('/api', apiRouter)` attaches a sub-router at a

Gotchas

`r.Mount('/api', apiRouter)` attaches a sub-router at a prefix. `apiRouter` is itself a `*chi.Mux`. Mount strips the prefix before passing the request to the sub-router. This enables fully independent route trees for modular codebases.

A9 `r.NotFound(handler)` sets a custom 404 handler for

Assessment

`r.NotFound(handler)` sets a custom 404 handler for unmatched routes. `r.MethodNotAllowed(handler)` sets a 405 handler when a route exists but not for the used HTTP method. Without custom handlers, Chi returns plain text 404/405 by default.

A5 Use standard context: `ctx := context.WithValue(r.Context(),`

Contextualization

Use standard context: `ctx := context.WithValue(r.Context(), userKey{}, user); r = r.WithContext(ctx)`. Retrieve in the handler: `user := r.Context().Value(userKey{}).(User)`. Chi recommends unexported struct keys to avoid collisions with other packages.

A10 Chi handlers are plain `http.HandlerFunc`, so use

Optimization

Chi handlers are plain `http.HandlerFunc`, so use `httptest.NewRequest` and `httptest.NewRecorder` directly. Set chi URL params with `chitest: rctx := chi.NewRouteContext(); rctx.URLParams.Add('id', '42'); r = r.WithContext(context.WithValue(r.Context(), chi.RouteCtxKey, rctx))`.